THE COKERNEL OF A MAP OF ADEQUATE FUNCTORS IS ADEQUATE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The cokernel of a map of adequate functors into a module-valued functor, and its embedding into a module-valued functor. Same chapter, Lemma lemma-cokernel-into-module, invoked at proof lines 696-703. Gives adequacy of the intermediate cokernels and the embeddings used at both stages.

Source reference. Original lemma 06V1: sources/stacks-adequate/pinned-original-source.tex, source lines 647–647. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Let $\varphi : F \to G$ be a map of adequate functors on Alg_A. Then* $\mathrm{Coker}(\varphi)$ *is adequate.*

Proof. Choose an injection $G \to \underline{M}$. Then we have an injection $G/F \to \underline{M}/F$. By Lemma 06V1 we see that $\underline{M}/F$ is adequate, hence we can find an injection $\underline{M}/F \to \underline{N}$. Composing we obtain an injection $G/F \to \underline{N}$. By Lemma 06V1 the cokernel of the induced map $G \to \underline{N}$ is adequate hence we can find an injection $\underline{N}/G \to \underline{K}$. Then $0 \to G/F \to \underline{N} \to \underline{K}$ is exact and we win. □

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